

TenantFlow Architecture Report

v0.8.0 Engineering Phase: Automated Tenant Provisioning & Flyway Migration Blueprint

STATUS: APPROVED

DOMAIN: Infrastructure & Core Platform

TARGET RELEASE: v0.8.0

1. EXECUTIVE SUMMARY

This blueprint details the structural transformation of the **TenantFlow** infrastructure to support **Zero-Touch Provisioning**. By automating physical schema generation and executing programmatic database migrations via Flyway, the system achieves instant tenant onboarding, eliminates manual database administration, and ensures standardized structural integrity across all isolated environments.

2. CURRENT VS. TARGET ARCHITECTURAL TOPOLOGY

The architectural transition shifts schema initialization from manual SQL administration to an API-driven, automated lifecycle workflow:

Metric / Feature	v0.7.0 (Discrete Schema Routing)	v0.8.0 (Automated Provisioning)
Schema Creation	Manual database administration	API-driven dynamic generation
Database Migration	Application startup (monolithic)	Programmatic execution (per tenant)
Onboarding Latency	High (Requires manual intervention)	Zero (Instantaneous REST API execution)
Structural Integrity	Prone to human oversight	Guaranteed via versioned Flyway SQL scripts

3. CORE IMPLEMENTATION STRATEGY

To achieve dynamic provisioning without restarting the application, TenantFlow will leverage the Flyway Java API integrated within a centralized service layer:

Step 1: Dynamic Schema Generation

Implement a centralized provisioning service to intercept registration payloads. This component executes native PostgreSQL `CREATE SCHEMA` commands dynamically upon verifying tenant uniqueness.

Step 2: Programmatic Flyway Execution

Bypass standard Spring Boot auto-configuration for Flyway. Instantiate Flyway programmatically targeting the newly created schema context, executing versioned DDL scripts (e.g., `v1__init_tables.sql`) to construct foundational enterprise tables instantly.

4. NEXT IMPLEMENTATION MILESTONES

- Implement Master Admin CRUD API endpoints for managing the lifecycle, suspension, and deletion of provisioned tenants.
- Integrate centralized audit logging to track all automated DDL executions and schema modifications during the onboarding phase.